T3 - TEARCI
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Questions and answers
GROUP 1 - IR Transmitter

1- **Operation and Frequency control of 555**
The 555 IC works in astable mode. I use the formula for frequency: $f = 1.44 / ((R1 + 2R2) * C1)$. For duty cycle: $D = (R2 / (R1 + 2R2)) * 100\%$. R1 goes from 15kΩ min to 25kΩ max, keeping duty around 13%. The components that control the frequency are R1, R2 and C1 (they set the charge and discharge times).
F(hz)Rmin=4700 F(hz)Rmin=3150

2- **Infrared LED Current Behaviour**
I use 13% duty cycle so the LED doesn't overheat and saves battery. Peak current formula: $I_{peak} = (V_{supply} - V_{led}) / R_{led}$. Average current: $I_{avg} = I_{peak} * (\text{duty cycle} / 100)$, so it's only 13% of peak when on, and zero when off.
Duty 13%
$I_{peak}=0{,}117$
$I_{avg}=0{,}015$

3-**Role of the BD140 Transistor**
The BD140 is a PNP transistor with beta (hFE) usually 40 to 160. Beta formula: $\beta = I_c / I_b$. It acts as a switch: small base current from 555 controls bigger collector current to the LED.
I_b=25ma
beta=4,68

4-**Purpose of Capacitor C2**
C2 is like a small battery. It stores energy with formula $Q = C * V$, and releases it to keep current steady, so no drops in voltage for the LED.

GROUP 2 - IR Receiver

1-**Photodiode Behaviour (Light Sensor)**
The photodiode detects the infrared beam by generating a photocurrent proportional to the light intensity. It is connected in reverse bias to increase its speed and sensitivity. This configuration widens the depletion region, reducing junction capacitance for a faster response to the 4.5 kHz pulse. When the beam is blocked, the photocurrent drops to its minimum (dark current), causing a voltage change that triggers the alarm sequence.

2-**Transistor Amplifier Stage (Signal Conditioning)**
The two NPN transistors (T1 and T2) amplify the weak photocurrent signal into a robust voltage signal for the PLL. Two stages are used to achieve sufficient voltage gain and better isolation. They are operated "slightly into saturation" to intentionally over-drive the signal. This squares the waveform to a roughly 50% duty cycle pulse train, which is a clean, stable signal easily recognized by the NE567 PLL.

3- **Phase-Locked Loop (NE/LM567) Function**
The PLL acts as a highly selective tone decoder that detects the 4.5 kHz pulsed frequency from the transmitter. When the PLL is locked onto the signal (beam intact), its output at Pin 8 goes Low. This Low state indicates the presence of the valid signal. When the beam is interrupted, the signal disappears, the PLL loses lock, and Pin 8 goes High, which is the signal used to activate the alarm.

formula : f = 1/(1.1*Rt*Ct) (Rt,Ct, pin 5 and 6 resistor and capacitor)

Rt = 10k Ct = 22n

f = 4,13Khz

4- **Alarm Oscillator (NE555) Role**
The NE555 (IC2) is configured as an astable multivibrator to generate an audible alarm tone. It is activated by the PLL's output (IC1 Pin 8) connected to the Reset pin (Pin 4) of IC2. When the beam is intact, Pin 8 is Low, holding IC2 in reset. When the beam is interrupted, Pin 8 goes High, releasing the reset and allowing the oscillator to start.

formula 1 : f = 1.44/((R1+2R2)C) (R1, R2, C, components between the 5v and pin 7,6,2 and gnd)
R1 = 1k
R2 = 100K
C = 47n

f = 152hz

formula 2 : R2 = ((1.44/(400*c)) - R1)/2

R2 = ~40K
f = ~400hz